

Abhishek Murti

Bioinformatician | Data Scientist | Computational Biologist
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Professional Summary

Computational biologist with expertise in large-scale multi-omics data analysis and statistical modeling. Proven track record of building scalable bioinformatics pipelines, developing analysis frameworks, and contributing to high-impact publications while reducing operational costs and enabling precision medicine breakthroughs. Experienced in leading cross-institutional collaborations and passionate about using AI-driven approaches to accelerate biological discovery and generate testable, data-driven hypotheses.

Experience

University of California, San Francisco — Bioinformatician

Dec 2021 – Present

- Analyzed large-scale **single-cell**, **single-nuclei**, and **spatial transcriptomics** datasets, with over **1M+ cells**, using Seurat, Scanpy, and Squidpy to investigate immune-mediated diseases and tumor microenvironments.
- Identified tumor signatures for risk stratification in **20 hepatoblastoma patients** by developing parallel single-cell and single-nuclei RNA sequencing pipelines, **expanding cohort size by 40%** through inclusion of archived biobanked samples and enabling **cross-platform validation** for precision medicine.
- Designed a **graph-based algorithm** for **tumor-region assignment** in spatial transcriptomics, validated across **31 TMA cores** spanning multiple hepatoblastoma subtypes—including cases with **multiple subtypes within the same core**—and integrated with scRNA-seq and snRNA-seq to analyze tumor-immune interactions.
- Implemented and validated SNP-based consensus **demultiplexing algorithms** for single-nuclei sequencing workflows, allowing **66% reduction in sequencing costs** through **improved sample multiplexing efficiency**.
- Led autoimmune hepatitis study integrating bulk, single-nuclei RNA sequencing and spatial transcriptomics data from **8 patient biopsies**, uncovering a **novel autoantigen** and a **therapeutically targetable** immune pathway from minimal tissue input.
- Contributed to multi-organ, cross-institutional atlas projects (**COVID Tissue Atlas** and **Tabula Sapiens 2.0**) encompassing **85,000+ cells** from **20+ human organs**, spearheading the liver data analysis to identify systemic disease responses and establish reference cell-type profiles for translational research.

Selected Technical Projects

– AI Scientist Framework

Developed an end-to-end Python and LangChain pipeline leveraging **LLMs** to classify hepatoblastoma subtypes from single-cell transcriptomics, predict risk profiles, and generate biologically interpretable hypotheses with supporting literature, **streamlining data-driven hypothesis generation** for experimental follow-up.

– Julia Transcriptomics Toolkit

Created `scAM.jl`, an open-source Julia package for NGS data analysis with complete workflows from preprocessing through clustering and differential expression, **implementing 15+ core functions (UMAP, QC, visualization)** to accelerate analyses and provide a **high-performance alternative** to existing R/Python tools.

Education

Cornell University, NY — MS

August 2019 - July 2021

Chemical and Biomolecular Engineering, GPA: 3.768/4

Thesis: Mechanistic Modeling of a Cell-Free Glucose Biosensor for Therapeutic Applications

Birla Institute of Technology and Science, Pilani, India — BE

August 2015 - July 2019

Major: Chemical Engineering, Minor: Finance, GPA: 9.21/10

Technical Skills

Programming Languages: Python (Scanpy, Pandas, NumPy), R (Seurat, tidyverse, ggplot2), Bash, Julia, MATLAB
Data Analysis & Bioinformatics: NGS, Single-cell & Single-nuclei RNA-seq, Spatial Transcriptomics, Clustering, Dimensionality reduction (UMAP, PCA), Pathway Analysis, Gene Ontology, GWAS
Workflow & Cloud: High Performance Computing (HPC), Docker, Git, Google Cloud Platform, Nextflow
Visualization & Reporting: Matplotlib, Seaborn, Plotly

Publications

- Kim, J.J., Kurial, S.N., Choksi, P.K., Nunez, M., Lunow-Luke, T., Bartel, J., Driscoll, J., Her, C.L., Dhillon, S., Yue, W. and **Murti, A.**, 2025. AAV capsid prioritization in normal and steatotic human livers maintained by machine perfusion. *Nature biotechnology*, pp.1-13.
- Granados, A.A., Bucher, S., Song, H., Agrawal, A., Chen, A.T., Peng, T., Neff, N., Pisco, A.O., Huang, F., Wang, B. and **COVID Tissue Atlas Consortium**, 2023. Single-nuclei characterization of pervasive transcriptional signatures across organs in response to COVID-19. *Elife*, 12, p.e81090.
- Liu, J., Tran, V., Vemuri, V.N.P., Byrne, A., Borja, M., Kim, Y.J., Agarwal, S., Wang, R., Awayan, K., **Murti, A.** and Taychameekiatchai, A., 2023. Concordance of MERFISH spatial transcriptomics with bulk and single-cell RNA sequencing. *Life science alliance*, 6(1).
- Alkhani, A., Baskaran, S., **Murti, A.**, Rapp, B., Levy, C.S., Wang, B. and Nijagal, A., 2024. Perinatal liver inflammation is associated with persistent elevation of CXCL10 and its canonical receptor CXCR3 on common myeloid progenitors. *bioRxiv*, pp.2024-08.
- Quake, S.R. and **Tabula Sapiens Consortium**, 2024. Tabula Sapiens reveals transcription factor expression, senescence effects, and sex-specific features in cell types from 28 human organs and tissues. *bioRxiv*, pp.2024-12.
- Adhikari, A., **Murti, A.**, Narayanan, A.M., Lim, H.E. and Varner, J.D., 2023. Modeling and Analysis of a Cell-Free Gluconate Responsive Biosensor. *bioRxiv*, pp.2023-01.

Posters & Presentations

- **Murti, A.**, et al. “Effects of Intravenous Heme and Placebo on Gene Expression in Patients with Acute Hepatic Porphyrin Attacks” **Oral presentation and poster** presented at: *International Porphyrin Symposium*; October 2025 *Awarded one of the 3 best talks.*
- **Murti, A.**, et al. “Single-Cell and Single-Nuclei Transcriptomics Reveal Hepatoblastoma Tumor Signatures for Precision Oncology” **Poster** presented at: *5th International Pediatric Liver Tumor Research Meeting*; June 2025.
- **Murti, A.**, et al. “Single-Cell and Single-Nuclei Transcriptomics Reveal Hepatoblastoma Tumor Signatures for Precision Oncology” **Oral presentation** at: *UCSF Annual Liver Center Symposium*; May 2025.
- **Murti, A.**, et al. “A single nuclei RNA sequencing approach to characterize autoimmune hepatitis biology” **Oral presentation** at: *UCSF Annual Liver Center Symposium*; May 2024.
- **Murti, A.**, et al. “Harnessing the Power of Next Generation Sequencing to Understand Liver Pathophysiology” **Poster** presented at: *UCSF Annual Liver Center Symposium*; May 2023.